

Francisco Miranda

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EDUCATION

- 2024 - 2026 MSc in Systems, Control & Robotics at **KTH** (Stockholm) (GPA: 4.8/5.0)
2021 - 2024 BSc in Mechanical Engineering at **IST** (Lisbon) (GPA: 18/20)
Awarded annual Academic Excellence Award (3 consecutive years), Best Mechanical Engineering Student Award, and Sustainability Award for Bachelor's Final Project

RELEVANT EXPERIENCE

- Ericsson AB — MSc Thesis Research Intern** Jan 2026 - present
– Researching improvements of current state-of-the-art 3D edge-assisted mapping and visual localization pipelines, with applications in XR and vision-based robotic systems.
- SMaRC (KTH) — Research Intern** Jan 2025 - Dez 2025
– Investigated search planning algorithms for the ALARS autonomous drone system (SAAB-supervised), enabling reliable retrieval of unmanned underwater vehicles (UUVs).
– Integrated and benchmarked a hybrid YOLO–Canny edge detection pipeline for real-time UUV tracking, replacing an underperforming detector and establishing the project baseline method.
- TLMoto MotoStudent Team — Vehicle Dynamics Team Lead** May 2022 - Apr 2024
– Designed and validated longitudinal and lateral PID cascade controllers for the motorcycle simulation tool, enabling 4 new testing scenarios and weekly design iterations.
– Led a team of 7 students to optimize motorcycle geometry and deliver a complete suspension system design on schedule for competition.

PROJECTS

- HaGIR (Hand Gesture Instructed Robot) — Personal Project** [Link to Project](#)
Designed and trained a custom CNN for finger count detection (public + self-collected data), achieving 93% accuracy; derived the DC motor dynamic model via system identification and implemented the motor control system.
- Robotics and Autonomous Systems — Project Course** [Link to Project](#)
Deployed inverse kinematics, behaviour trees, and localization (ICP) algorithms on a ROS2-based robotic system to explore unknown workspaces and sort detected objects, achieving 100% pick-and-place accuracy and outperforming all other project teams.

SKILLS

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|-----------------------------------|---|
| Programming Languages & Libraries | Python, C++, MATLAB, PyTorch, OpenCV, NumPy |
| Frameworks & Tools | ROS2, Git, Linux, Docker, Kubernetes |
| Languages | Portuguese (Native), English (C1), Spanish (A2) |

PAPERS

- AUV2026 Gone Fishing: Post-Mission Autonomous Aerial Recovery of Unmanned Underwater Vehicles (Co-Author, accepted)